Fractional-Domain Information-Enhanced Hyperspherical Prototype Learning Method for Hyperspectral Image Open-Set Classification

Shou Feng, *Member, IEEE*, Shengnan Wang, Congan Xu, Chunhui Zhao, Wei Li, *Senior Member, IEEE*, and Ran Tao, *Senior Member, IEEE*

***Abstract*— In recent years, research in the field of hyperspectral image classification (HSIC) has increasingly focused on the open-set problem. Open-set classification (OSC) demands not only accurately classifying the known categories but also identifying the unknown samples that are not labeled or included within the training data during the testing stage. Existing open-set methods often suffer from misclassification between the known and unknown categories due to their inadequate utilization of metric space. Moreover, relying on a single-threshold strategy performs poorly in identifying unknown categories in complex open environments. In this article, a fractional-domain information-enhanced hyperspherical prototype learning (FrHSPL) method is proposed for hyperspectral image (HSI) OSC. FrHSPL develops a hyperspherical prototype learning (HSPL) strategy that ensures the features of known categories are uniformly distributed on the hypersphere. Therefore, HSPL can effectively enhance the interclass separability and optimize the exploitation of metric space. Subsequently, to enhance the discrimination capability of spectral features, a frequency-spatial–spectral information aggregation (FSSIA) module is devised to deeply integrate fractional domain information with spatial and spectral information. Finally, an open-set recognition (OSR) module is designed to identify unknown categories by using the prototypes of each known category along with the corresponding prototype radii. Extensive experiments on four common HSI datasets indicate that the proposed FrHSPL exhibits superior performance in comparison with both closed-set and open-set methods.**

Received 6 March 2025; revised 23 April 2025; accepted 15 May 2025. Date of publication 20 May 2025; date of current version 3 June 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62471155, Grant 62371153, and Grant 62271499; in part by China Postdoctoral Science Foundation under Grant 2023M740265; and in part by the Fundamental Research Funds for the Central Universities under Grant 3072025ZN0803 and Grant 3072024LJ0804. *(Corresponding author: Congan Xu.)*

Shou Feng is with the College of Information and Communication Engineering and the Key Laboratory of Advanced Marine Communication and Information Technology, Ministry of Industry and Information Technology, Harbin Engineering University, Harbin 150001, China, and also with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: fengshou@hrbeu.edu.cn).

Shengnan Wang and Chunhui Zhao are with the College of Information and Communication Engineering and the Key Laboratory of Advanced Marine Communication and Information Technology, Ministry of Industry and Information Technology, Harbin Engineering University, Harbin 150001, China (e-mail: wangsn@hrbeu.edu.cn; zhaochunhui@hrbeu.edu.cn).

Congan Xu is with the Research Institute of Information Fusion, PLA Naval Aviation University, Yantai 264001, China (e-mail: xcatougao@163.com).

Wei Li and Ran Tao are with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: liw@bit.edu.cn; rantao@bit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3571606

***Index Terms*— Fractional Fourier domain, hyperspectral image (HSI), hyperspherical prototype learning (HSPL), open-set classification (OSC).**

I. Introduction

HYPERSPECTRAL images (HSIs) are enriched with extensive spectral and spatial information, capable of precisely revealing the unique fingerprint effects of different land cover features [1], [2]. HSI classification (HSIC) meticulously analyzes the spectral information of each pixel within the imagery and assigns an accurate category label to each pixel [3]. As a pivotal means of ground objects interpretation, HSIC has demonstrated significant application value in various fields, including coastal wetland monitoring [4], [5], precision agriculture [6], [7], and forestry resource management [8], [9].

Given the significant high-dimensionality and complex nonlinear characteristics of HSI data [10], the extraction of effective features has emerged as a pivotal factor determining classification performance. In early explorations, traditional classifiers represented by support vector machine (SVM) [11], [12], k-nearest neighbor (KNN) [13] and the Bayesian estimation method [14] were used to directly classify each pixel of HSIs. Despite achieving competitive results, these traditional machine learning-based methods are prone to extract shallow features, which constrains their performance when confronted with varied hyperspectral scenarios.

In recent years, the integration of deep learning (DL) into the field of remote sensing has led to groundbreaking advancements in HSIC [15], [16]. Convolutional neural network (CNN), renowned for its ability in capturing spatial local information, has continually inspired numerous advanced methodologies [17], [18]. To address the limitations of CNN in capturing long-range dependencies, researchers have explored alternative backbone networks, such as graph convolutional network (GCN) [19], [20] and Transformer [21], [22], to model global relationships and long-distance dependencies. Meanwhile, given that the features extracted by deep neural networks are often excessively complex, various attention modules have been ingeniously incorporated into the networks [23], [24], aiming to focus on critical information and enhance feature representation.

Due to environmental variations or sensor noise, the spectral differences among various categories of hyperspectral

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

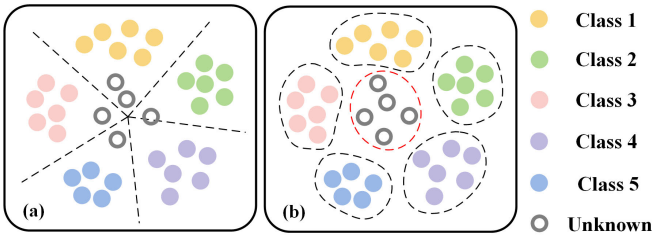


Fig. 1. Schematic of closed-set and open-set classifications. (a) CSC. (b) OSC (the circles of different colors denote the known categories, while the annuluses denote the unknown categories).

data often become indistinguishable [25]. For instance, the spectral curves of graminoids and cattails in wetland scenarios exhibit high similarity [26], [27], while the spectral curves of the same crop at different growth stages also show variations in agriculture [28]. These two forms of spectral uncertainty greatly impede the complete representation of the inherent properties of land cover, thereby affecting the precise delineation of classification decision boundaries. To enhance the discriminability between spectral signatures of different land covers, researchers have begun to explore the utilization of frequency-domain information as an auxiliary [29]. Zhao et al. [30] explored a novel image Transformer (FrIT) for multimodal remote sensing image classification. Zhang et al. [31] employed the fractional Fourier transform (FrFT) to improve the performance of hyperspectral anomaly detection. Therefore, leveraging the advantages of frequency-domain information in distinguishing subtle spectral differences among land cover holds significant potential for further enhancing the model performance.

The existing advanced HSIC models have showcased substantial advancement in performance, but they are predominantly designed based on the static closed-set assumption [32], [33]. As a result, they may be unable to effectively identify samples belonging to unknown categories. In practical applications, classification systems are typically deployed in open environments, where testing sets may contain labels of unknown categories [34]. The closed-set classification (CSC) methods employing linear classification layers or softmax function tend to misclassify pixels of unknown categories into known ones, as shown in Fig. 1(a), leading to over-determination of the target land cover [35]. A straightforward improvement involves setting a threshold for the model's prediction confidence, with pixels below this threshold being deemed as unknown [36]. However, the effectiveness of this method heavily relies on the setting of a rigid threshold, and pixels with low-classification confidence do not always belong to unknown categories [37]. Consequently, current CSC methods remain inadequate in effectively recognizing land cover of unknown categories in open-set environments. To overcome the limitations of the CSC and better adapt to open environments, the open-set recognition (OSR) task was first proposed by paper [38]. Open-set classification (OSC) demands the accurate classification of known categories while identifying and labeling samples that do not belong to these known categories as unknown, as depicted in Fig. 1(b). OSC offers greater practical value due to its ability to identify all unknown samples as an unknown category, in contrast to CSC.

Currently, research on OSC for HSIs is relatively scarce and can be roughly divided into two types of approaches: methods based on the reconstruction theory and those that integrate multiple learning strategies [39]. The former relies on reconstruction error to identify whether input belongs to unknown categories, typically assuming that unknown samples will induce higher reconstruction error [40], [41]. Consequently, Liu et al. [42] introduced a multitask DL method (MDL4OW), which reconstructs images and utilizes extreme value theory to label pixels that cannot be effectively reconstructed as unknown categories. To further leverage the spectral–spatial features of samples, a spectral–spatial latent reconstruction learning (SSLR) method is proposed for detecting unknown categories [43]. To alleviate the dependence on labeled samples, a self-supervised multitask learning (SSMTL) framework is proposed, which identifies unknown categories by analyzing the uncertainty of predicted probability and reconstruction loss [44]. Although methods based on reconstruction show certain potential in addressing unknown categories, they also require manual threshold setting, which limits the recognition of unknown categories and undermines the adaptability of these methods.

In comparison, selecting an appropriate learning strategy may be more suitable for OSC tasks. Through integrating an improved generative adversarial network with evidential theory, Ji et al. [45] introduced a spectral–spatial evidential learning (SSEL) network to balance the classification performance for both known and unknown categories. Chen et al. [28] proposed a fractional frequency-spatial–spectral prototype network (FrFSSPN), which applied the FrFT for OSC in agricultural hyperspectral imagery, thereby improving the recognition accuracy for unknown categories. Recently, some researchers introduced a reciprocal points learning strategy to minimize the overlap in the feature space between known and unknown categories [46]. These methods are effective in identifying unknown categories across various scenarios.

However, the above methods generally underutilize the metric space, resulting in insufficient separation between known categories. This increases the risk of misclassification between known and unknown categories. Moreover, existing OSR methods often rely on a single threshold to identify unknown categories. However, they fail to fully exploit and leverage the distributional characteristics of known categories, thereby limiting their ability to effectively discriminate unknown categories. In addition, the instability of model classification performance caused by spectral variations remains an unavoidable challenge in open-set HSIC tasks.

To address the aforementioned problems, a novel fractional-domain information-enhanced hyperspherical prototype learning (FrHSPL) method is proposed in this article. To improve the exploitation of the metric space, a hyperspherical prototype learning (HSPL) strategy is proposed to ensure that features of known categories are uniformly distributed on a hypersphere. Therefore, HSPL can effectively enhance interclass separability while keeping compact intraclass

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

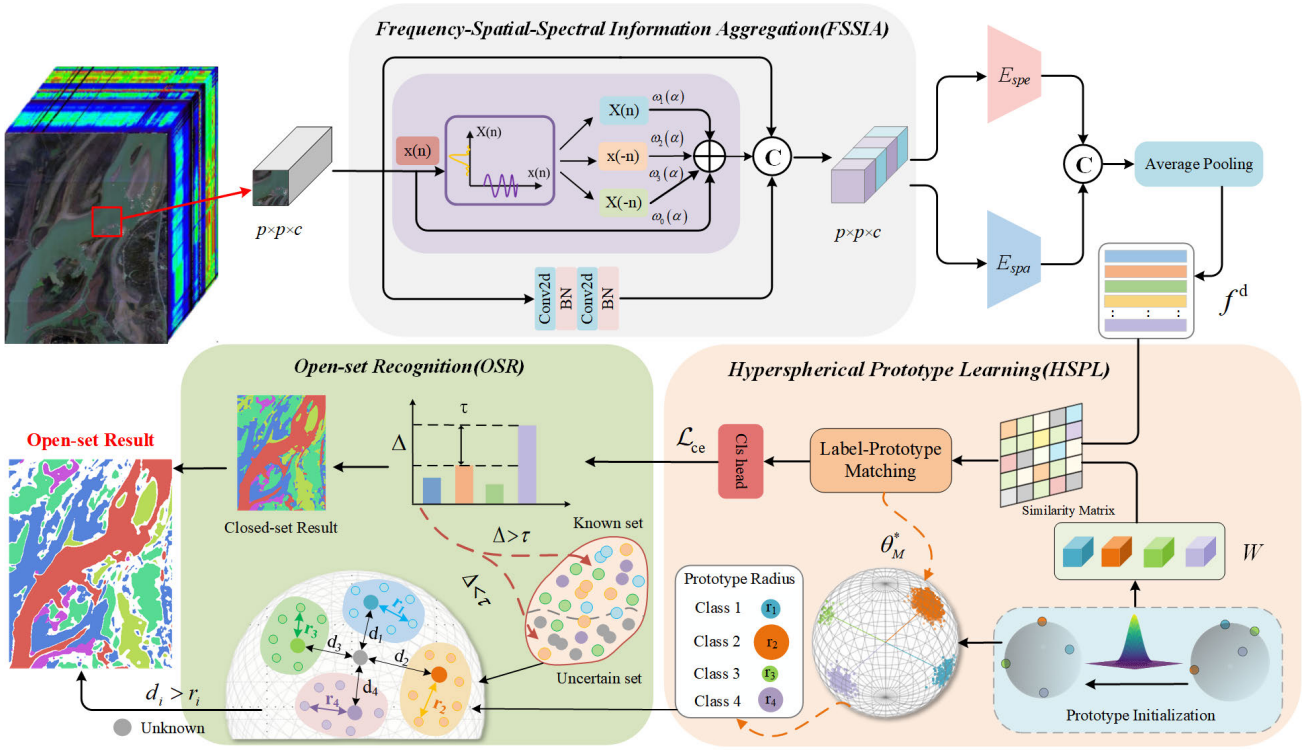


Fig. 2. Overall structural schematic of the proposed FrHSPL. Initially, training data are fed into the FSSIA and further processed by a spectral feature extraction network (E_{spe}) and a spatial feature extraction network (E_{spa}). Then, the HSPL strategy is applied to enhance the network's discriminative ability for features of known categories. Finally, an OSR module, based on the prototypes of known categories and their prototype radii, is designed to identify unknown categories.

relationships. Next, the weighted FrFT (WFrFT) is introduced to aggregate the fractional domain information and the conventional spatial–spectral domain information, thereby alleviating the challenge of indistinguishable spectral data differences. Finally, we also designed an OSR module based on each known category prototype and prototype radius, avoiding the reliance solely on softmax classification probability or a single threshold for identifying unknown categories. The main contributions of this article can be summarized as follows.

1) We proposed an FrHSPL method for open-set HSIC. Through dynamically matching known category labels with uniformly distributed hyperspherical prototypes, the HSPL strategy effectively exploits the metric space in a manner consistent with interclass relationships. Samples of the same category are more tightly clustered, whereas those from different categories are distinctly separated, thereby enhancing the model's classification performance.
2) To enhance the discriminability of spectral features and consequently improve the separability among different known categories, a frequency-spatial–spectral information aggregation (FSSIA) module is devised to deeply aggregate fractional domain information with spatial–spectral information. We also designed a dual-branch feature extraction network, ensuring the efficient utilization of frequency-spatial–spectral information.
3) To overcome the limitation of identifying unknown categories solely through a single threshold, we designed an OSR module, which is grounded on the prototypes of each known category and their prototype radii. During the inference stage, samples in the uncertain set with Euclidean distances from known prototypes exceeding their respective prototype radii are then identified as unknown categories. Fully exploring and leveraging the distribution characteristics of known categories can help improve the model's ability to identify unknown categories.

The remainder of this article is organized as follows. In Section II, the proposed FrHSPL is presented in detail. Section III presents and analyzes the experimental results. Finally, the conclusion is provided in Section IV.

II. Methodology

The overall framework of FrHSPL is illustrated in Fig. 2. First, the training data are fed into an FSSIA module to integrate the fractional domain and the spatial–spectral domain information. The integrated data is subsequently processed by a dual-branch feature extraction network to obtain the final feature of the data. Next, the HSPL strategy is applied to enhance the network's discriminative ability for features of known categories. Finally, all testing data are sent to the model to derive predicted probability and feature data in the inference stage, and the unknown categories are identified by the designed OSR.

A. FSSIA Module

Similar to common HSIC methods, FrHSPL takes patches of HSIs as input, and the central pixel label is treated as the label

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

for the entire patch. Specifically, let $X \in \mathbb{R}^{p\times p\times c}$ be an input patch, where $p \times p$ and c denote the patch size and number of bands, respectively. Given a training set $\mathcal{D}_{\mathcal{L}} = \{X_1, \ldots, X_{n_s}\}$ containing n_s labeled samples that cover N known categories, where $y_i \in \{1, \ldots, N\}$ corresponds to the label of the input X_i. Unlike the CSC, the OSC task needs to address unknown categories during the inference stage. Therefore, the labels for the testing data $\mathcal{D}_{\mathcal{T}} = \{t_1, \ldots, t_m\}$ are $\{1, \ldots, N, N+1\}$, where $y_j \in \{N+1\}$ typically represents the label for unknown categories.

To obtain the fractional domain information of the data, the WFrFT is applied along the spectral dimension. Previous studies have demonstrated the excellence of the FrFT in HSIs processing [28]. However, FrFSSPN relies solely on a single FrFT, resulting in its sensitivity to fractional order selection. As an extension of FrFT, the WFrFT can more effectively capture crucial frequency information by weighting different frequency components [47], offering greater flexibility. Meanwhile, the weighting mechanism of the WFrFT can be regarded as a soft attention mechanism applied to the projections of spectral features in different fractional domains, enabling the extraction of more discriminative features in complex scenarios.

The WFrFT of sample X_i is expressed as follows:

$$f_i^{\alpha} = \left(1/c\right) \sum_{n=1}^{c} X_i(n)\Psi(\alpha; u; n) \tag{1}$$

where Ψ is the kernel function of the α-order WFrFT and n is the channel index and must be an integer. The fractional domain information of the sample is represented as f_i^{α} after the α-order WFrFT is performed. The definition of the kernel function is as follows:

$$\begin{aligned}\Psi(\alpha; u; n) = {} & \omega_0(\alpha)\delta(u-n) + \left(1/\sqrt{2\pi}\right)\omega_1(\alpha)\exp(-jun) \\ & + \omega_2(\alpha)\delta(u+n) + \left(1/\sqrt{2\pi}\right)\omega_3(\alpha)\exp(jun)\end{aligned} \tag{2}$$

where u denotes the fractional domain transformation plane and $\omega_l(\alpha)$ serves as the weighting coefficient for WFrFT, defined specifically as follows:

$$\omega_l(\alpha) = \left(1/4\right) \sum_{n=0}^{3} \exp\left[-j2\pi n(\alpha - l)/4\right] \tag{3}$$

where $l = 0, 1, 2, 3$, $j = (-1)^{1/2}$.

Furthermore, two convolutional layers are also employed to capture the local features of the input samples. The first layer employs 1×1 convolutional kernel to extract shallow spectral features of the data, while the second layer utilizes 3×3 convolutional kernel to capture shallow spatial features. To enhance the network's nonlinear expressive power, batch normalization and ReLU activation function are sequentially applied after each convolutional layer.

Finally, the raw input data, along with its fractional domain information and shallow spatial–spectral features, are integrated and fed into the dual-branch feature extraction network.

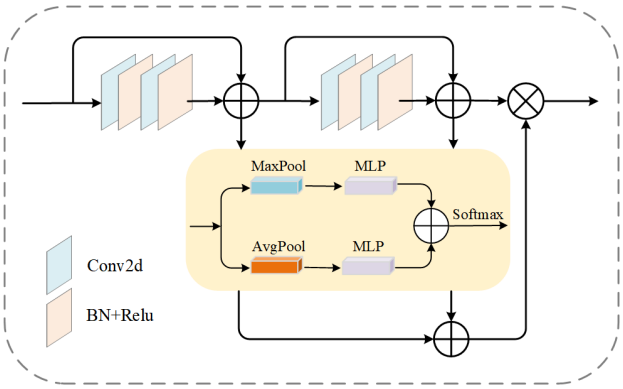


Fig. 3. Details of the spectral feature extraction network.

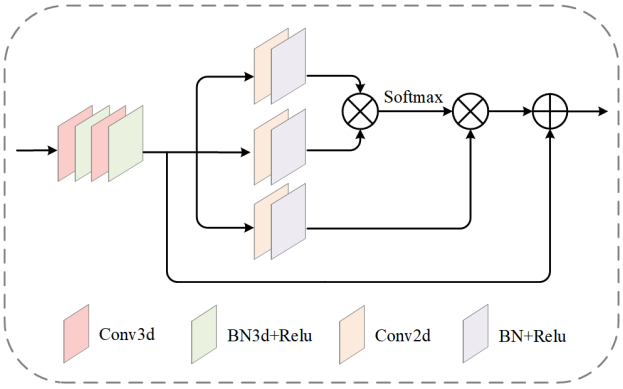


Fig. 4. Details of the spatial feature extraction network.

B. Dual-Branch Feature Extraction Network

To extract deep features containing richer class discrimination information, the data integrated through FSSIA are, respectively, fed into spectral and spatial feature extraction networks.

1) Spectral Feature Extraction Network: The spectral feature extraction network (E_{spe}), as depicted in Fig. 3, consists of two spectral residual blocks (SRBs) and a channel attention module (CAM). Specifically, the input first passes through multiple Conv2D layers to capture frequency-spectral features at various depths, with SRBs capable of preserving low-level feature information. Then, the outputs of the two SRBs are individually sent into the CAM, and the attention weights outputted by the CAM are utilized to weight the original channel features, thereby highlighting the contribution of key channels. To enhance the network's multilevel feature representation capability, the attention weights obtained from different depth features are summed and multiplied with the network's output features, ultimately yielding enhanced frequency-spectral features.

2) Spatial Feature Extraction Network: The spatial feature extraction network (E_{spa}) comprises two Conv3D layers and a multiscale attention module (SAM), as illustrated in Fig. 4. First, two Conv3D layers are applied to preliminarily extract local frequency-spatial features from the data, effectively capturing local correlations across both spatial and spectral dimensions. Then, the SAM is designed with three parallel branches, each employing a convolutional kernel of different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

sizes, to explicitly capture features across varying receptive fields. By leveraging Softmax to compute the importance weights of features across different scales, enabling the model to focus on spatial regions that are crucial for classification tasks. Finally, the fusion of the original features with the output of the SAM preserves the global information of the original features. At the same time, it also incorporates the local details from SAM, resulting in a more effective feature representation.

The frequency-spectral and frequency-spatial features extracted from the dual-branch feature extraction network are concatenated to obtain the output features $\boldsymbol{f}^d$ of the backbone network, where d denotes the feature dimension.

C. HSPL Strategy

To optimize the exploitation of metric space, the HSPL is proposed to constrain the feature distributions of known categories onto a hypersphere. By leveraging the geometric properties of the hypersphere, HSPL can effectively maintain interclass separability while enhancing intraclass compactness. In addition, HSPL also simplifies the decision boundaries between different known categories, making the classification results more accurate.

The HSPL initializes N equally spaced points on a hypersphere as hyperspherical prototypes, in a manner that is independent of both the training data and the network architecture. The problem of uniformly distributing N prototypes on the unit hypersphere in the case of $d = 2$ can be simplified to dividing the unit circle into N equal slices with $(2\pi/N)$ angles.

However, the optimal uniform distribution solution does not exist for $d \geq 3$ [48]. Therefore, the Gaussian potential kernel is introduced to estimate the generally optimal distribution of prototypes on the hypersphere. The Gaussian potential kernel $G_t(u, v) \triangleq S^d \times S^d \to \mathbb{R}_+$ can be defined as follows:

$$G_t(u, v) \triangleq e^{-t\| u-v\|_2^2}, \quad t > 0 \tag{4}$$

where t is the bandwidth parameter, S^d represents the d-dimensional unit hypersphere, and P_w signifies the distribution of prototypes on this hypersphere. Then, the uniformity loss can be represented as follows:

$$\mathcal{L}_{\text{un}} \triangleq \log \mathop{\mathbb{E}}_{w_i, w_j \overset{\text{i.i.d.}}{\sim} P_w} [G_t(u, v)], \quad t > 0 \tag{5}$$

where w_i and w_j are the prototype vectors within the prototype matrix $\boldsymbol{W} \in \mathbb{R}^{d\times N}$ and $\boldsymbol{W}$ is randomly initialized during the first iteration of training.

The uniformity loss $\mathcal{L}_{\text{un}}$ is closely associated with the uniform distribution of points on a hypersphere. Notably, the solution for the distribution of prototypes on the hypersphere is independent of the backbone and classifier, but solely depends on d of the metric space and the number N of classes. Upon deriving the solution of W, HSPL adopts a dynamic matching strategy based on label-to-prototype alignment to align the known category information with the corresponding hyperspherical prototypes. The mapping from the output feature $\boldsymbol{f}_i$ to its assigned can be learned through the following formula:

$$\mathcal{L}_{\text{sp}} = \sum_{i=1}^{n_s} \frac{1}{2} \left(f_i^\top \boldsymbol{W} \theta_M^*(y_i) - 1 \right)^2 \tag{6}$$

where θ_M^* denotes a bijective mapping from labels to prototypes. This mapping oversees intraclass compactness by reducing the similarity between sample feature $\boldsymbol{f}_i$ and its corresponding prototype $\boldsymbol{W}\theta_M^*(y_i)$. Considering that the hyperspherical prototypes remain fixed and unupdated throughout the training stage, the primary objective is to ascertain a mapping, as detailed in the following equation:

$$\theta_M^* = \arg\min_{\theta_M} \sum_{i=1}^{N} \left[-\overline{f}_i^\top \boldsymbol{W} \theta_M(i) + \log \sum_{j=1}^{N} e^{\overline{f}_j^\top \boldsymbol{W} \theta_M(j)} \right] \tag{7}$$

where $\overline{\boldsymbol{f}}_i$ is the momentum averaging over the feature representations of the ith class. Here, a momentum update method is adopted to determine this mapping [49].

The aforementioned mapping aligns sample features with class prototypes by maximizing the log-likelihood function of observed classes, thereby achieving a uniform distribution of known category features on the unit hypersphere. Ideally, the combination of $\mathcal{L}_{\text{sp}}$ and fixed, equally distributed hyperspherical prototypes can achieve optimal interclass separation while enforcing intraclass compactness.

In addition, to ensure the performance of the closed-set classifier, the cross-entropy loss is calculated as follows:

$$\mathcal{L}_{\text{ce}} = -\frac{1}{n_s} \sum_{i=1}^{n_s} \sum_{j=1}^{N} y_{i,j} \log(p_{i,j}) \tag{8}$$

where $y_{i,j}$ denotes the true label indicating that sample i belongs to the jth known class, while $\boldsymbol{p}_{i,j}$ represents the

Algorithm 1 FrHSPL

Initialize θ_M^*, $\boldsymbol{W}$, $T_1 > 0$ and $T_2 > 0$.
for $t = 0, 1, \ldots, T_1 - 1$ **do**
  Compute $\mathcal{L}_{un}$, $\forall \boldsymbol{w}_i \in \boldsymbol{W}$;
  Update $\boldsymbol{W}$ using $\nabla_{\boldsymbol{W}} \mathcal{L}_{un}$;
  Save $\boldsymbol{W}$; ▷ prototype initialization
end for
for $t = 0, 1, \ldots, T_2 - 1$ **do**
  for data in $\mathcal{D}_{\mathcal{L}}$ **do**
    $\boldsymbol{f}^d = E(data)$
    Compute $\mathcal{L}_{sp}$ using $\boldsymbol{f}^d$, θ_M^*, and $\boldsymbol{W}$;
    Compute $\mathcal{L}_{ce}$ using $\boldsymbol{f}^d$, y_i;
    Update E_θ using Eq.(9); ▷ training
  end for
  Save prototype set P and prototype radius set R;
end for
Input: HSI data $\mathcal{D}_{\mathcal{T}} = \{t_1, \ldots, t_m\}$ for testing;
Output: The open-set predicted result $\hat{y}$;
Predict the closed-set label $\hat{y} = \arg\max(E(t_i))$;
if $\hat{y} \in Uncertain_set$ **then**
  Calculate the Euclidean distance $d_{\hat{y}}$ between $\boldsymbol{f}_i$ and $\boldsymbol{p}_{\hat{y}}$;
  if $d_{\hat{y}} > r_{\hat{y}}$ **then**
    $\hat{y} = unknown$; ▷ testing
  end if
  return $\hat{y}$
end if

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

probability predicted by the classifier, signifying the likelihood of sample i belonging to the jth class.

Consequently, the gross loss function of the entire network should combine two functions to minimize the loss jointly

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{sp}}. \tag{9}$$

In the context of supporting classification loss, $\mathcal{L}_{\text{sp}}$ not only obviates the need to recalculate class prototypes during each training iteration but also facilitates the derivation of more compact and representative prototypes for known categories. Algorithm 1 provides a detailed presentation of the complete execution procedure of FrHSPL, where E denotes the dual-branch network.

D. OSR Module

To address the limitations of relying solely on a fixed threshold for unknown category recognition, a novel OSR module is devised. The cornerstone of this module lies in leveraging the prototypes of known categories along with their dynamically adjusted prototype radii. Through adaptively learning the prototype radius for each known category, the feature distribution of known categories can be represented more finely, thereby enhancing the accuracy of OSR tasks.

The average of all feature vectors within the same category is determined to be the prototype, yielding a prototype set $P = \{\boldsymbol{p}_1, \ldots, \boldsymbol{p}_N\}$. In addition, the mean Euclidean distance from samples within a category to its designated prototype is calculated, defining this as the prototype radius. As a result, a radius set $R = \{r_1, \ldots, r_N\}$ is obtained

$$r_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \sqrt{\left(\boldsymbol{f}_{i,j} - \boldsymbol{p}_i\right)^2}, \qquad i \in \{1, \ldots, N\} \tag{10}$$

where $\boldsymbol{f}_{i,j}$ represents the jth sample feature of class i and n_i is the number of samples in class i.

To mitigate the risk of misclassifying known categories as unknown, the identity of test samples is initially inferred based on closed-set prediction probabilities. OSR is then applied exclusively to samples whose prediction results are deemed uncertain. Given the uncertain nature of the logit vector distribution for unknown categories, we avoid relying directly on the magnitude of logit values for judgment. Instead, this uncertainty is quantified by calculating the difference between the maximum and second-maximum value $\Delta(\cdot)$ within the closed-set probability. For a given test sample t_i

$$t_i \in \begin{cases} \text{Known} \quad \text{set}, & \Delta(t_i) > \tau \\ \text{Uncertain} \quad \text{set}, & \Delta(t_i) \leq \tau \end{cases} \tag{11}$$

where τ is set to 0.8 as a hyperparameter in this article to enforce a rigorous initial criterion.

Samples with the substantial uncertainty in predicted probabilities are subsequently grouped into an uncertainty set. This identity inference based on predicted probability not only allows for preliminary identification of unknown categories but also ensures accurate classification of known categories. Next, the feature vector $\boldsymbol{f}_i$ of the sample within the uncertainty set is matched with the result label obtained from the CSC.

TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR SALINAS DATASET

No.	Land cover type	Train	Test
1	Weeds-1	40	1969
2	Weeds-2	40	3686
3	Fallow	40	1936
4	Fallow-P	40	1354
5	Fallow-S	40	2638
6	Stubble	40	3919
7	Celery	40	3539
8	Grapes	40	11231
9	Soil	40	6163
10	Corn	40	3238
11	Lettuce-4wk	40	1028
12	Lettuce-5wk	40	1887
13	Lettuce-6wk	40	876
14	Lettuce-7wk	40	1030
15	Vinyard-U	40	7228
16	Vinyard-T	40	1767
17	Unknown	0	5613
	Total	640	59102

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR DTL DATASET

No.	Land cover type	Train	Test
1	Lake Water	40	1080
2	Tidal Flat	40	696
3	Meadow	40	837
4	Reed	40	966
5	Willow	40	604
6	Unknown	0	433
	Total	200	4616

The Euclidean distance between $\boldsymbol{f}_i$ and the corresponding class prototype $\boldsymbol{p}_i$ is then calculated and denoted by d_i. If the value of d_i exceeds the prototype radius r_i of the respective category, it indicates low similarity between the sample and the predicted known categories. In such cases, the pixel is identified as belonging to the unknown categories.

III. EXPERIMENT AND RESULTS ANALYSIS

In this section, four hyperspectral experimental datasets, and the specific details of the experimental setup are first introduced. Then, the classification performance of the proposed FrHSPL is evaluated and compared with seven related algorithms. Furthermore, to verify the effectiveness of the devised modules, ablation experiments are conducted, and discussions and analyses are provided regarding the key model parameters that may influence the method's performance.

A. Experimental Datasets

1) Salinas: The Salinas dataset [50] was captured by the AVIRIS spectrometer in Salinas Valley, which has a 3.7-m

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR LN02 DATASET

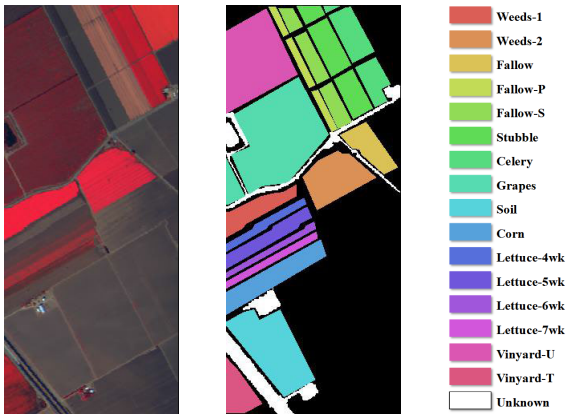
No.	Land cover type	Train	Test
1	Liao River	40	12706
2	Aquaculture Ponds	40	56568
3	Paddy Fields	40	36988
4	Intertidal Muds	40	26304
5	Phragmites Australis	40	70293
6	Suaeda Salsa	40	7505
7	Construction Land	40	4033
8	Reed Pond	40	6920
9	Unknown	0	1132
	Total	320	222449

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR KSC DATASET

No.	Land cover type	Train	Test
1	Scrub	40	721
2	Willow swamp	40	203
3	CP hammock	40	216
4	Slash pine	40	212
5	Oak/Broadleaf	40	121
6	Hardwood	40	189
7	Swamp	40	65
8	Graminoid marsh	40	391
9	Cattail marsh	40	364
10	Salt marsh	40	379
11	Mud flats	40	463
12	Water	40	887
13	Unknown	0	520
	Total	480	4731

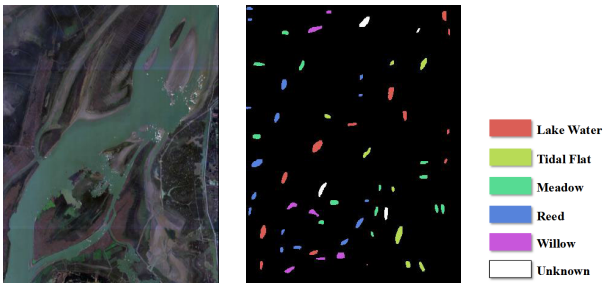
spatial resolution and a wavelength scope of 0.4–2.5 μm. It contains 512 × 217 pixels and 224 bands after pruning the noisy channels. The closed-set dataset of Salinas contains a total of 16 known land cover types. The noncrop features, including buildings, roads, and ponds, are manually labeled as the unknown category in order to be clearly distinguished from crops. Details about the Salinas dataset are presented in Table I, while the false-color image and open-set ground truth are shown in Fig. 5.

2) Dongting Lake (DTL): The DTL dataset [51] was acquired by GF-5 satellite, over the Dongting Lake basin, Hunan. It contains 456 × 352 pixels with a spatial resolution of 30 m, and the spectral wavelength cover ranges from 0.4 to 2.5 μm. It retains 310 spectral bands after removing 20 water absorption spectral bands that contain noise. The DTL dataset includes a total of six land cover types, with Carex, whose growth cycles are significantly influenced by seasonal fluctuations, specifically labeled as an unknown category. Table II presents the details of the training and testing samples. Fig. 6 shows the false-color image and open-set ground truth of the DTL dataset.



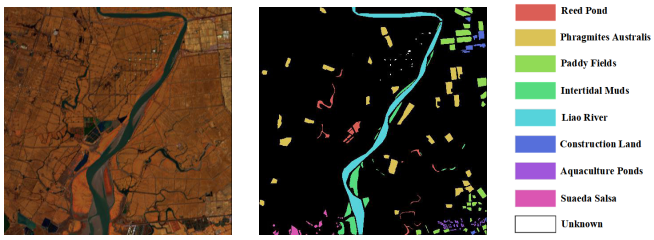
(a) (b) (c)

Fig. 5. Overview of the Salinas dataset. (a) False color image. (b) Ground truth. (c) Land cover type.



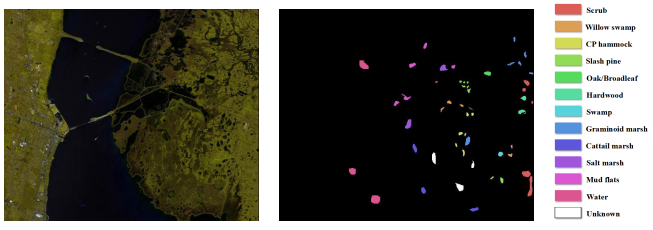
(a) (b) (c)

Fig. 6. Overview of the DTL dataset. (a) False color image. (b) Ground truth. (c) Land cover type.



(a) (b) (c)

Fig. 7. Overview of the LN02 dataset. (a) False color image. (b) Ground truth. (c) Land cover type.

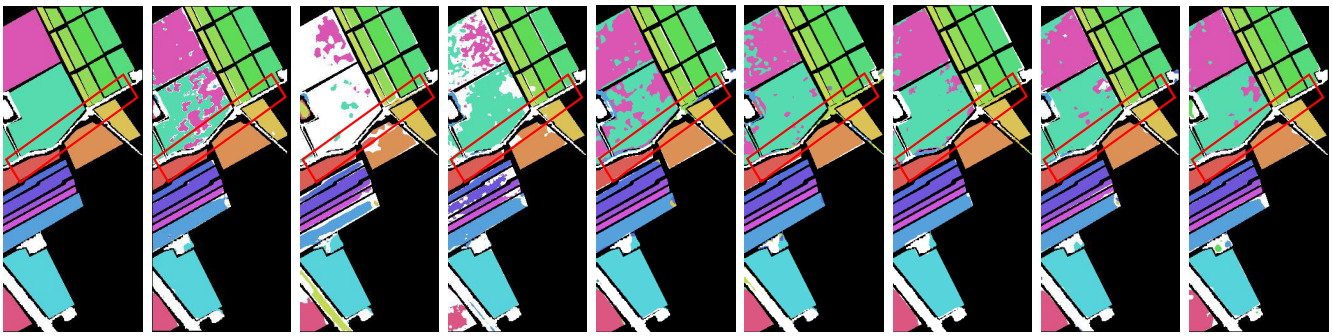


(a) (b) (c)

Fig. 8. Overview of the KSC dataset. (a) False color image. (b) Ground truth. (c) Land cover type.

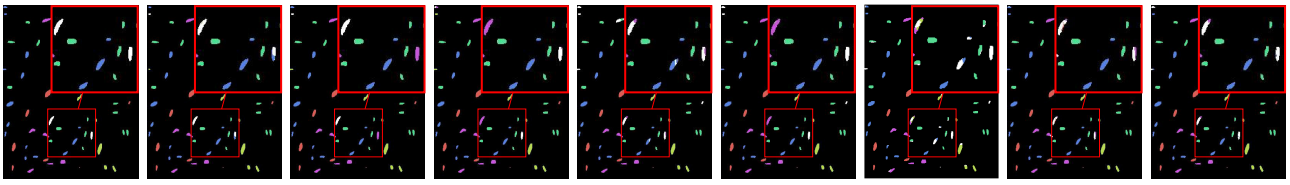
3) LN02: The LN02 dataset [52] was captured by the ZY-1(02D) satellite at the Liaohe River Estuary Wetland in Panjin, Liaoning, China, with coordinates at 121°46.41.E and 40°56.17′N. The spatial size of the dataset is 512 × 512 pixels

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 9. Classification maps of different methods on the Salinas dataset. (a) Ground truth. (b) SSRN (84.68% OA). (c) DBDA (60.36% OA). (d) SSFTT (78.98% OA). (e) MDL4OW (88.08% OA). (f) SSLR (90.80% OA). (g) SSMTL (93.58% OA). (h) FrFSSPN (94.95% OA). (i) FrHSPL (95.24% OA).



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 10. Classification maps of different methods on the DTL dataset. (a) Ground truth. (b) SSRN (92.44% OA). (c) DBDA (93.88% OA). (d) SSFTT (93.35% OA). (e) MDL4OW (92.12% OA). (f) SSLR (94.19% OA). (g) SSMTL (88.68% OA). (h) FrFSSPN (96.46% OA). (i) FrHSPL (97.69% OA).

with a spatial resolution of 30 m and mainly covers the areas along the Liaohe River. After removing bad bands, it contains 144 bands from the spectral range of 0.4–2.5 μm. The LN02 dataset includes nine land cover types, the industrial land which exhibits significant differences from wetland crops is labeled as an unknown category. This can also facilitate the supervision and management of wetland resources. The number of samples for each class is listed in Table III, while the false-color image and open-set ground truth are shown in Fig. 7.

4) KSC: The KSC dataset [53] was captured by AVIRIS sensor over Florida with spectral coverage from 0.4 to 2.5 μm. With a spatial resolution of 18 m, the KSC dataset encompasses a range of 614 $\times$ 512 pixels. After removing water absorption and low-SNR bands, it contains 176 bands. This dataset includes a total of 13 land cover types, with Spartina marsh, which poses a threat to coastal zone ecology, being labeled as an unknown category. Details about the Salinas dataset are presented in Table IV, while the false-color image and open-set ground truth are shown in Fig. 8.

B. Experimental Details

The proposed FrHSPL is implemented within Python3.7 leveraging the PyTorch framework. All parameters of the FrHSPL network are randomly initialized, with patch size set to 9. To enhance training efficiency, the batch size is adjusted to 512, and the number of training epochs is set to 200. In addition, the order α of the WFrFT employs an adaptive strategy for determination, while the probability threshold τ is set to 0.8. These parameter choices will be elaborated in Section III-E. It is important to note that our algorithm optimization is achieved through two independent stages. First, obtaining fixed and uniformly distributed prototype points on a hypersphere using (5). Then, optimizing (9) to solve for the optimal model parameters.

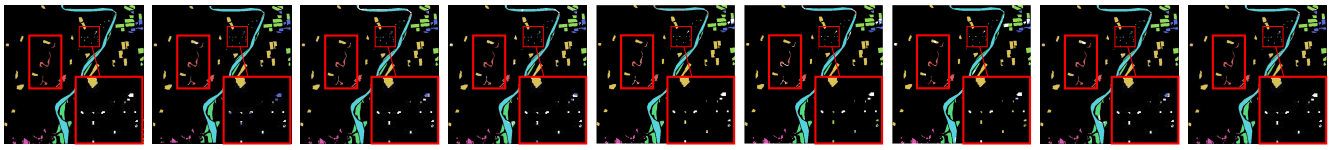
All experiments were conducted on the same hardware platform, equipped with a GTX-3090 GPU, an Intel 4210R CPU, and 40 GB of RAM, and repeated ten times under identical settings to obtain average results and the standard deviation. Finally, the overall accuracy (OA), average accuracy (AA), and the Kappa coefficient are used to evaluate the classification performance [1].

C. Experiments and Analysis

To verify the performance of the proposed FrHSPL for open-set HSIC tasks, seven influential methods are selected for comparison. Primarily, FrHSPL is evaluated against cutting-edge OSC methods, including FrFSSPN [28], MDL4OW [42], SSLR [43], and SSMTL [44]. In addition, to validate the superiority of open-set methods in handling unknown categories, FrHSPL is also compared with conventional CSC methods. The two CNN-based methods, spectral–spatial residual network (SSRN) [17] and dual-branch dual-attention (DBDA) network [24], as well as a Transformer-based method, spectral–spatial feature tokenization Transformer (SSFTT) [21], are chosen.

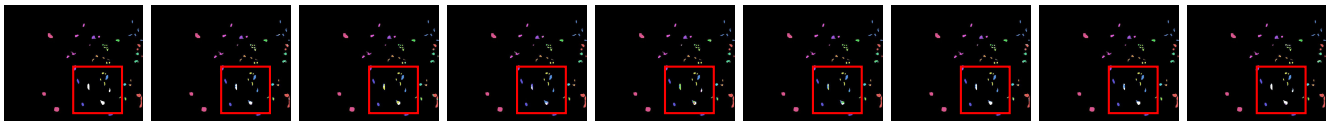
Furthermore, following the general solution for open-set problems [42], [54], the confidence threshold for the closed-set models is set to 0.5, where pixels with a maximum prediction probability below 0.5 will be taken as the unknown category. In [55], the maximum value of the softmax probability vector is referred to as the maximum softmax probability (MSP) baseline, and its effectiveness has been empirically validated across multiple benchmark datasets. To ensure fairness, all comparative methods are configured according to their optimal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 11. Classification maps of different methods on the LN02 dataset. (a) Ground truth. (b) SSRN (93.80% OA). (c) DBDA (92.48% OA). (d) SSFTT (93.05% OA). (e) MDL4OW (93.83% OA). (f) SSLR (94.56% OA). (g) SSMTL (93.24% OA). (h) FrFSSPN (93.89% OA). (i) FrHSPL (95.03% OA).



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 12. Classification maps of different methods on the KSC dataset. (a) Ground truth. (b) SSRN (88.46% OA). (c) DBDA (89.41% OA). (d) SSFTT (89.12% OA). (e) MDL4OW (92.84% OA). (f) SSLR (91.34% OA). (g) SSMTL (88.97% OA). (h) FrFSSPN (93.15% OA). (i) FrHSPL (95.68% OA).

TABLE V

CLASSIFICATION ACCURACIES OF METHODS ON THE SALINAS DATASET

Class	SSRN [17]	SSFTT [21]	DBDA [24]	FrFSSPN [28]	MDL4OW [42]	SSLR [43]	SSMTL [44]	**FrHSPL**
1	93.73±0.07	97.19±1.65	95.79±4.48	**99.76±0.36**	89.78±0.38	95.86±0.68	99.55±0.28	98.45±1.34
2	**99.57±0.24**	95.61±3.05	94.83±6.18	99.52±0.30	94.39±0.32	97.92±0.50	96.64±2.97	97.38±1.56
3	99.37±0.40	98.07±0.88	83.63±5.49	98.49±2.99	99.43±0.40	**99.81±0.24**	96.73±2.34	98.56±0.80
4	98.32±0.11	97.94±1.18	93.91±1.64	98.58±0.60	98.51±0.49	98.62±1.02	95.78±4.00	**99.54±0.73**
5	97.81±1.62	93.23±3.45	84.64±6.44	97.55±0.55	91.38±3.70	97.82±1.02	93.14±1.06	**98.57±1.71**
6	97.44±0.19	97.84±0.62	96.69±1.67	99.09±0.84	98.61±0.73	**99.97±0.03**	97.08±3.53	98.78±1.01
7	97.45±0.50	94.77±1.99	94.66±4.04	98.32±0.38	95.16±0.41	98.22±0.87	89.67±4.95	**98.62±0.83**
8	63.58±8.16	47.71±7.92	18.25±5.54	91.08±2.97	69.71±4.73	87.87±5.19	87.71±3.38	**91.84±0.41**
9	98.36±0.88	99.17±0.33	98.81±2.21	99.15±0.41	98.21±0.47	**99.74±0.22**	96.68±1.61	97.93±1.05
10	90.26±2.72	91.11±1.75	68.69±6.61	91.33±4.84	93.36±0.80	**98.68±0.69**	92.42±2.95	91.34±1.95
11	95.09±0.69	95.89±1.86	78.41±4.67	**99.86±0.26**	94.98±0.63	99.60±0.42	97.89±1.99	98.98±1.04
12	96.02±0.23	81.82±7.67	88.76±6.93	**99.98±0.17**	99.96±0.06	99.90±0.11	99.81±0.27	99.55±0.50
13	96.65±0.08	85.58±7.13	93.47±3.41	97.57±2.65	99.12±0.81	99.22±0.24	99.03±1.05	**99.36±0.65**
14	93.42±2.44	85.58±7.13	89.17±5.17	97.92±2.57	98.22±0.28	98.60±0.22	98.09±0.95	**98.78±0.95**
15	82.82±5.76	48.46±7.41	20.33±5.89	91.22±3.20	82.57±3.98	85.08±3.42	93.25±1.24	**93.37±1.75**
16	76.76±3.46	64.83±8.62	84.09±4.10	97.97±1.41	97.25±0.33	**98.45±0.39**	93.43±3.65	95.34±1.81
Unknown	80.69±3.50	89.87±0.57	61.98±2.44	87.63±1.84	80.86±1.07	55.58±4.75	85.72±2.99	**91.56±1.03**
OA(%)	84.68±2.47	78.98±2.26	60.36±2.19	94.95±0.88	88.08±1.01	90.80±0.84	93.58±1.55	**95.24±0.16**
AA(%)	90.92±1.67	86.08±1.95	74.79±3.52	96.56±0.61	93.03±0.38	92.31±0.66	95.43±1.65	**96.88±0.11**
Kappa	83.31±2.77	77.06±2.34	57.12±2.36	94.76±1.21	89.52±1.74	90.20±1.44	93.88±1.66	**94.89±0.15**

settings reported in their original literature and trained using 40 training samples per known class.

1) Experimental Results on Salinas Dataset: Table V summarizes the quantitative results of all methods on the Salinas dataset. Under the same experimental conditions, FrHSPL achieves the most outstanding classification performance, with OA as high as 95.24%, AA of 96.88%, and Kappa coefficient of 94.89%. The classification accuracy for unknown classes reaches 91.56%, significantly surpassing those of other comparative methods. Furthermore, the standard deviation of FrHSPL experimental results is relatively small, indicating its stability. Notably, FrHSPL also leverages fractional domain information to enhance the discriminative capability of spectral features, effectively distinguishing the same crop at different growth stages (Lettuce-4wk, Lettuce-5wk, Lettuce-6wk, and Lettuce-7wk) within known categories.

The qualitative classification results are presented in Fig. 9. Some open-set comparison methods incorrectly classify the unknown categories within the red frame as various known classes, whereas FrHSPL demonstrates an ability to accurately identify unknown categories and exhibits clear classification boundaries. Due to the spectral feature confusion at the edges of ground objects, FrHSPL also misclassifies some pixels as unknown, resulting in suboptimal accuracy for certain known categories. However, in practical application scenarios, these minor errors have a negligible impact on the overall analysis. Overall, FrHSPL has exhibited exceptional classification performance on the Salinas dataset.

2) Experimental Results on DTL Dataset: Table VI provides the quantitative results of all methods on the DTL dataset. Evaluated on the OA, AA, and Kappa coefficient metrics, FrHSPL outperforms all comparative methods, leading the suboptimal method by 1.23%, 1.8%, and 1.3%, respectively. Notably, FrHSPL achieves a remarkable recognition accuracy of 91.13% for unknown categories, achieving an average improvement of 27.88% compared with other methods.

Given the balanced category distribution and significant differences among known categories in the DTL dataset, all methods maintain a decent level of classification accuracy, with OA consistently above 90%. However, closed-set comparisons misclassify unknown categories into different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CLASSIFICATION ACCURACIES OF METHODS ON THE DTL DATASET

Class	SSRN [17]	SSFTT [21]	DBDA [24]	FrFSSPN [28]	MDL4OW [42]	SSLR [43]	SSMTL [44]	FrHSPL
1	99.78±0.26	99.41±0.17	99.83±0.19	98.64±1.92	94.34±0.83	95.53±0.69	94.47±0.58	**99.89±0.14**
2	99.87±0.48	99.90±0.05	**99.91±0.14**	99.45±0.18	93.80±1.01	99.04±0.68	85.33±6.63	99.39±1.06
3	99.93±0.13	**99.98±0.02**	99.90±0.18	98.40±2.51	97.15±0.76	99.70±0.59	92.01±3.82	99.45±0.83
4	94.31±1.12	94.47±3.36	96.79±1.03	**97.47±1.51**	95.44±0.61	97.19±3.09	91.44±1.80	96.22±1.01
5	97.79±2.75	99.71±0.58	**99.97±0.46**	95.74±4.95	94.70±0.69	97.32±0.18	93.88±5.11	96.47±3.14
Unknown	55.38±5.52	42.18±4.65	40.92±5.15	84.97±4.86	72.98±0.83	71.24±3.73	75.05±5.98	**91.13±1.25**
OA(%)	92.44±3.92	93.35±0.25	93.88±0.89	96.46±1.15	92.12±0.52	94.19±0.93	88.68±1.00	**97.69±0.31**
AA(%)	91.21±1.68	88.92±0.81	89.43±1.64	95.41±1.06	91.35±0.41	92.80±1.24	86.51±1.28	**97.21±1.51**
Kappa	93.21±1.13	91.85±0.31	92.51±1.10	95.89±1.29	93.34±0.49	92.93±1.13	91.79±0.98	**97.19±1.07**

TABLE VII
CLASSIFICATION ACCURACIES OF METHODS ON THE LN02 DATASET

Class	SSRN [17]	SSFTT [21]	DBDA [24]	FrFSSPN [28]	MDL4OW [42]	SSLR [43]	SSMTL [44]	FrHSPL
1	92.10±3.99	83.37±4.15	81.30±3.20	90.56±2.44	77.90±2.66	83.74±6.92	91.63±3.92	**94.34±1.89**
2	98.16±0.67	95.61±2.08	98.83±0.98	97.61±1.26	99.43±0.28	**99.83±0.14**	93.20±2.66	95.83±2.90
3	88.15±0.74	89.73±2.05	91.77±3.77	92.21±3.25	**96.83±1.11**	96.52±0.30	89.85±3.54	93.05±1.73
4	**94.63±0.40**	91.74±2.01	89.23±3.24	92.27±1.02	93.24±0.59	91.95±2.13	87.62±6.64	93.88±2.54
5	98.27±0.59	95.37±1.18	91.42±3.25	95.28±2.21	95.02±2.52	95.29±1.39	95.16±1.95	**99.14±0.64**
6	88.92±1.69	94.17±1.33	94.31±2.55	93.57±2.33	66.11±2.17	78.52±1.13	76.91±6.62	**94.55±3.05**
7	94.82±0.87	93.64±2.78	92.01±1.87	93.98±2.08	89.76±2.01	90.74±2.76	91.64±4.79	**94.96±1.88**
8	73.73±0.72	91.42±1.26	90.73±1.63	91.29±0.82	97.35±0.66	97.53±0.47	**97.59±0.85**	92.19±2.21
Unknown	57.11±3.51	60.13±3.07	67.76±4.28	86.68±3.54	76.11±4.21	52.51±3.92	55.14±6.15	**89.56±1.02**
OA(%)	93.80±0.14	93.05±0.48	92.48±1.59	93.89±0.67	93.83±0.65	94.56±0.68	93.24±2.27	**95.03±0.42**
AA(%)	87.83±0.90	88.40±1.22	88.61±2.00	92.87±0.51	87.75±0.74	87.23±2.05	88.58±1.66	**94.43±0.37**
Kappa	93.22±0.46	91.31±0.58	91.24±1.80	93.07±1.38	92.32±0.64	93.90±1.24	93.16±2.36	**94.75±0.90**

TABLE VIII
CLASSIFICATION ACCURACIES OF METHODS ON THE KSC DATASET

Class	SSRN [17]	SSFTT [21]	DBDA [24]	FrFSSPN [28]	MDL4OW [42]	SSLR [43]	SSMTL [44]	FrHSPL
1	84.65±1.51	**99.57±0.55**	97.05±4.86	99.37±0.18	98.11±1.23	96.37±1.51	84.29±3.37	95.08±3.04
2	97.03±2.83	98.29±1.30	93.66±8.96	**99.91±0.16**	89.13±0.76	97.28±2.39	96.04±2.79	97.15±1.11
3	95.47±7.67	88.98±8.36	91.16±7.05	99.06±0.39	96.09±3.35	98.67±1.49	94.55±3.54	**99.12±1.19**
4	75.72±6.80	84.31±8.26	87.24±7.18	**99.92±0.15**	89.81±1.64	95.20±0.55	77.25±6.85	96.03±4.25
5	89.83±5.61	87.45±9.52	94.11±9.35	99.30±0.24	**99.87±1.01**	86.04±3.77	85.98±8.10	**90.47±4.80**
6	89.92±4.89	84.89±7.78	96.33±3.04	**99.91±0.17**	99.23±1.52	94.39±2.90	95.11±4.66	96.25±4.61
7	99.04±1.90	97.54±4.49	97.61±4.76	99.43±0.15	99.58±0.51	99.23±0.93	99.04±1.90	**99.92±0.76**
8	93.64±1.12	99.39±0.57	99.72±0.55	99.95±0.09	99.75±0.49	99.25±0.96	94.57±1.01	**100±0.00**
9	99.20±0.87	96.63±2.83	99.74±0.50	99.95±0.04	97.52±1.78	99.81±0.27	99.85±0.19	**99.97±0.09**
10	99.71±0.18	99.85±0.11	99.42±0.83	98.08±1.89	**99.93±0.13**	99.25±0.99	98.94±1.07	99.28±0.78
11	98.17±1.12	91.49±5.32	**99.84±0.31**	99.24±1.41	89.70±0.38	99.65±0.57	98.01±1.49	98.09±1.74
12	99.59±0.49	99.43±0.38	99.71±0.57	99.87±0.04	99.96±0.06	92.42±1.79	99.75±0.0.49	**100±0.00**
Unknown	62.69±4.74	30.28±3.17	28.54±3.04	44.69±3.76	23.90±3.80	15.64±5.14	63.57±3.73	**90.93±1.16**
OA(%)	88.46±2.86	89.12±2.34	89.41±1.80	93.15±1.24	92.84±0.48	91.34±1.67	88.97±2.04	**95.68±0.56**
AA(%)	89.02±2.68	89.45±3.46	88.75±3.39	94.64±0.95	90.12±0.35	86.01±1.31	88.85±2.91	**95.44±1.65**
Kappa	87.27±3.23	87.61±3.13	88.22±2.01	92.33±1.45	92.40±0.34	90.38±1.87	87.85±2.26	**95.63±1.54**

known categories, resulting in a recognition accuracy of only around 50% for unknown categories. Some OSC methods, on the other hand, misidentify certain known categories as unknown, thereby affecting the overall classification accuracy, as illustrated by the qualitative results in Fig. 10. In the DTL dataset, morphologically diverse and widely distributed Carex are labeled as unknown categories. Faced with these challenges, FrHSPL exploits the distribution characteristic of known categories, effectively identifying the unknown category. In addition, for known categories such as tidal flat and meadow with uneven spatial distribution, it can still maintain relatively high-classification accuracy.

3) Experimental Results on LN02 Dataset: Table VII presents the quantitative results of all methods on the LN02 dataset. Under identical experimental conditions, FrHSPL still exhibits optimal and stable classification performance, achieving OA, AA, and Kappa coefficients of 95.03%, 94.43%, and 94.75%, respectively. For unknown categories, FrHSPL attains a classification accuracy of 89.56%, which is also the highest among all comparisons.

It is noteworthy that the class distribution in the LN02 dataset is highly imbalanced, where the seventh class has 17 times fewer samples than the fifth class, and the number of unknown category samples being only 50th of that in the fifth class. Although the training samples for known categories are relatively balanced, this still poses significant challenges for metric space learning and unknown identification. Even under such unfavorable conditions, FrHSPL demonstrates exceptional performance in classifying each known class and identifying unknown categories, as illustrated by the qualita-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
ABLATION STUDY ON FOUR DATASETS

Module			Salinas			DTL			LN02			KSC		
FSSIA	HSPL	OSR	OA	Kappa	Unknown	OA	Kappa	Unknown	OA	Kappa	Unknown	OA	Kappa	Unknown
×	×	×	87.82%	86.66%	52.36%	90.31%	88.07%	36.56%	90.66%	89.34%	58.47%	88.57%	87.73%	37.69%
×	×	✓	89.89%	88.73%	65.17%	92.52%	90.88%	61.27%	92.69%	91.32%	77.17%	89.96%	88.85%	58.46%
✓	×	✓	92.53%	91.78%	84.55%	94.79%	93.65%	80.39%	93.77%	92.23%	81.92%	92.45%	91.62%	76.89%
×	✓	✓	93.28%	92.71%	87.47%	95.68%	94.73%	84.53%	93.81%	92.26%	83.89%	93.55%	93.26%	84.08%
✓	✓	×	92.42%	92.11%	71.34%	93.34%	91.82%	69.53%	93.61%	92.01%	68.64%	91.44%	90.37%	70.61%
✓	✓	✓	**95.24%**	**94.89%**	**91.56%**	**97.69%**	**97.19%**	**91.13%**	**95.03%**	**94.75%**	**89.56%**	**95.68%**	**95.63%**	**90.93%**

tive results in Fig. 11. Furthermore, for the known categories of Phragmites australis and Intertidal muds which have minor spectral differences, FrHSPL also provides more accurate and distinct classification results. This shows that FrHSPL can not only effectively exploit the metric space to address the uneven class distribution but also alleviate classification difficulties arising from these subtle spectral differences.

4) Experimental Results on KSC Dataset: Table VIII provides the classification results of various comparative methods on the KSC dataset and the classification maps are also depicted in Fig. 12. In comparison with OA and accuracy for the unknown category, the FrHSPL achieves the optimal overall performance with 95.68% and 90.93%, respectively. Notably, its accuracy for the unknown category outperforms other methods by an average of 45.90%.

The primary challenge in the KSC dataset lies in the spectral signature similarity among certain vegetation types, rendering the discrimination of land cover in this scenario particularly difficult. Conventional closed-set methods (e.g., DBDA and SSFTT) exhibit unstable performance and struggle to achieve satisfactory performance in identifying unknown categories. The situation becomes even more problematic when the spectral characteristics of unknown categories closely resemble those of known ones. In such cases, open-set methods that rely on a single threshold for unknown category recognition, such as MDL4OW and SSLR, suffer from significantly degraded capability. In comparison, FrHSPL has achieved optimal classification accuracy in nearly half of the known categories. Remarkably, for the Graminoid and Cattail, which exhibit highly similar spectral characteristics, FrHSPL also attains classification accuracies of 100% and 99.97%, respectively. Although spectral discrepancies are also addressed through the fusion of fractional-domain information by FrFSSPN, its classification accuracy for unknown categories merely reaches 44.69%. By employing a uniformly distributed hyperspherical prototype strategy, FrHSPL not only effectively enhances the feature discriminability of known categories but also significantly improves the recognition capability for unknown categories.

D. Ablation Study and Visualization

To rigorously validate the enhancement in model performance, an ablation study on three main modules is conducted in this section. In addition, a visualization comparison between the original data and the embedded features is provided.

1) Ablation Study: To further explore the effectiveness of different components, ablation studies are conducted on FSSIA, HSPL strategy, and OSR modules, respectively. The ablation results across four datasets are presented in Table IX. As indicated in the table, FrHSPL exhibits superior classification performance. Without the OSR module, the model solely relies on the backbone network for feature extraction, categorizing samples with a maximum classification probability below 0.5 as an unknown category. As evident from Table IX, the recognition accuracy for unknown categories is not satisfactory.

When the OSR module is applied independently, although the recognition accuracy for unknown categories improves, OA does not increase significantly. This also indicates that the performance of OSC methods is closely related to the model's effective extraction of features for known categories.

Notably, the integration of fractional domain information into the design of FSSIA has led to a significant enhancement in the model's classification performance. The OA is enhanced by 2.64%, 2.27%, 1.08%, and 2.49%, respectively, across the four datasets, and the recognition performance of unknown categories has also appreciably improved. This also highlights the advantages of fractional domain information in distinguishing subtle spectral differences of various land cover.

Furthermore, HSPL is employed to fully exploit the metric space, with an average improvement of 2.85% in OA. This enhancement convincingly illustrates that maximizing the separation among known categories reinforces the model's capacity for classifying. Moreover, it also yields an average increase of 19.48% in the recognition accuracy of unknown categories, optimizing the model's performance in identifying unknowns.

2) Visualization: To intuitively display the performance of HSPL in open-set feature space, the t-SNE dimension-reduction is utilized to project the features onto a 2-D plane and visualize both the original data and the embedded features. The visualization across four datasets is shown in Fig. 13. In this visualization, black represents the unknown category, while other colors denote various known categories. To ensure the precision of the data visualization, the number of feature points is identical to the number of test samples in the datasets.

As illustrated in Fig. 13(a)–(d), the distribution of original data is overlapping and disordered, with different known categories intertwined with each other and even samples within the same class failing to form compact clusters. Meanwhile, samples of unknown categories are widely distributed across the entire feature space, entangled with various

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

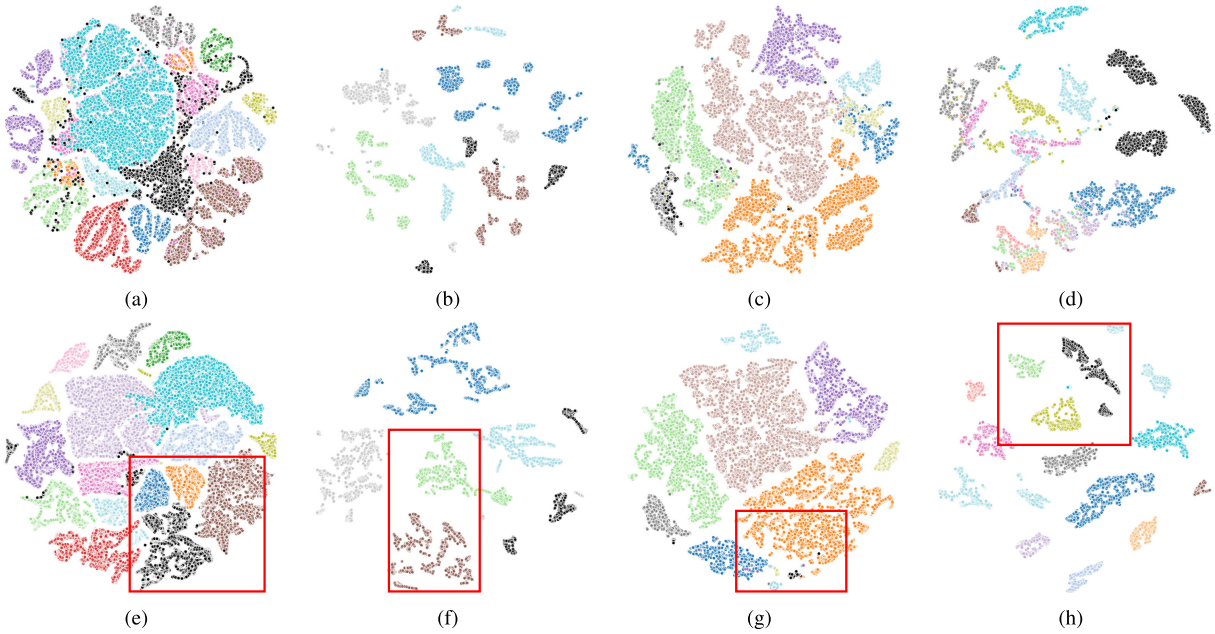


Fig. 13. Visualization projection of raw data versus HSPL embedding features. The embeddings of different categories are represented by different colors, with black denoting an unknown class. Raw data of (a) Salinas dataset, (b) DTL dataset, (c) LN02 dataset, and (d) KSC dataset. Embeddings of HSPL on (e) Salinas dataset, (f) DTL dataset, (g) LN02 dataset, and (h) KSC dataset.

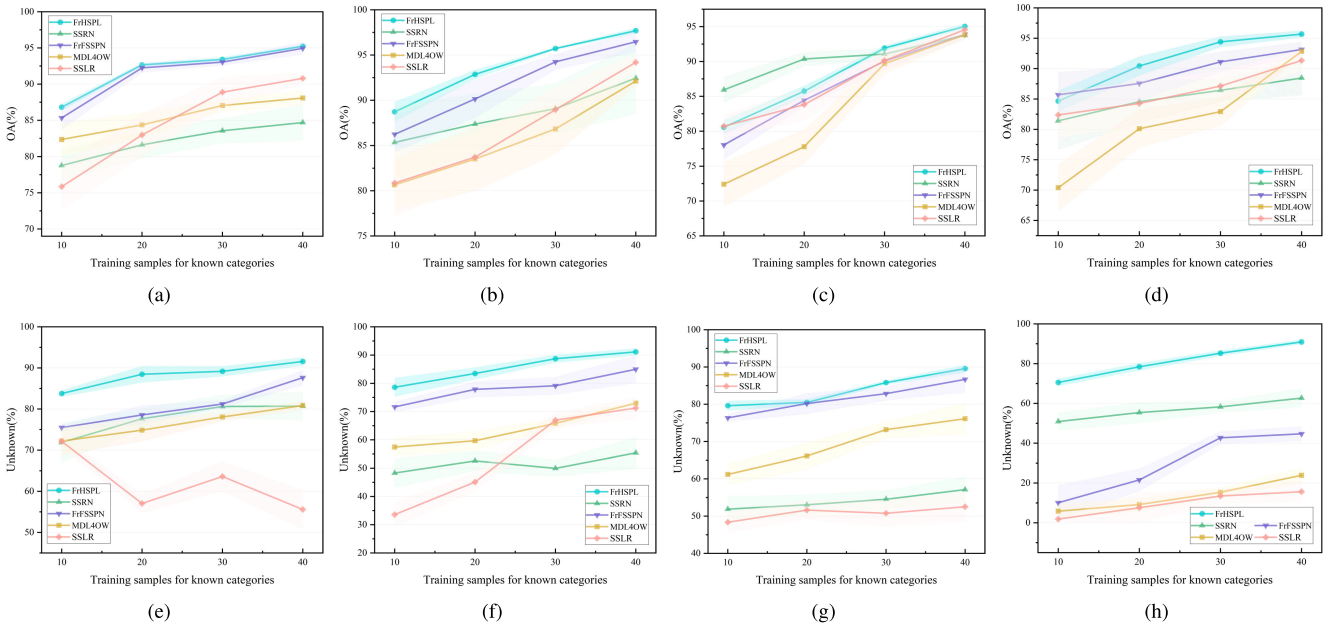


Fig. 14. Experimental results with varying numbers of training samples for known categories. OA for (a) Salinas dataset, (b) DTL dataset, (c) LN02 dataset, and (d) KSC dataset. Accuracy of unknown category on (e) Salinas dataset, (f) DTL dataset, (g) LN02 dataset, and (h) KSC dataset.

known categories. In contrast, features in the HSPL latent space exhibit greater interclass separation and more compact intraclass aggregation, as shown in Fig. 13(e)–(h). Different known categories are clearly separated from each other, while samples within the same class are tightly clustered together. Such distribution characteristics are conducive to improving the classification accuracy of known categories. Taking the DTL dataset as an example, the Meadow, colored in brown, appears relatively dispersed in its original data but significantly aggregated and well-separated from other categories when transformed into the HSPL feature space.

Furthermore, samples of unknown categories in the HSPL feature space are also distinguished from known categories to some degree and exhibit a centrally clustered pattern. In the LN02 dataset, the sample count of the unknown category differs significantly from that of the known categories, rendering it nearly indistinguishable from the original data. By comparison, samples of unknown categories within the HSPL open space can cluster together, as highlighted by the rectangular box in Fig. 13(g). This also facilitates the identification of unknown categories without compromising the classification accuracy of known categories.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

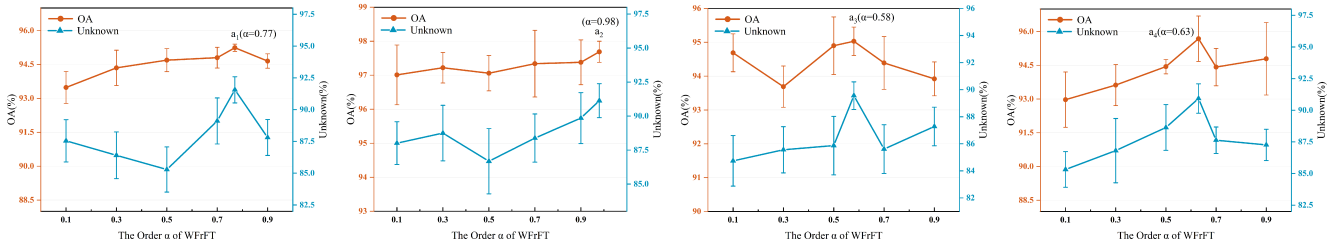


Fig. 15. Experimental results of WFrFT order α on four datasets, where a_1–a_4 represent the optimal model performance achieved by self-adaptation of order α. (a) Salinas dataset. (b) DTL dataset. (c) LN02 dataset. (d) KSC dataset.

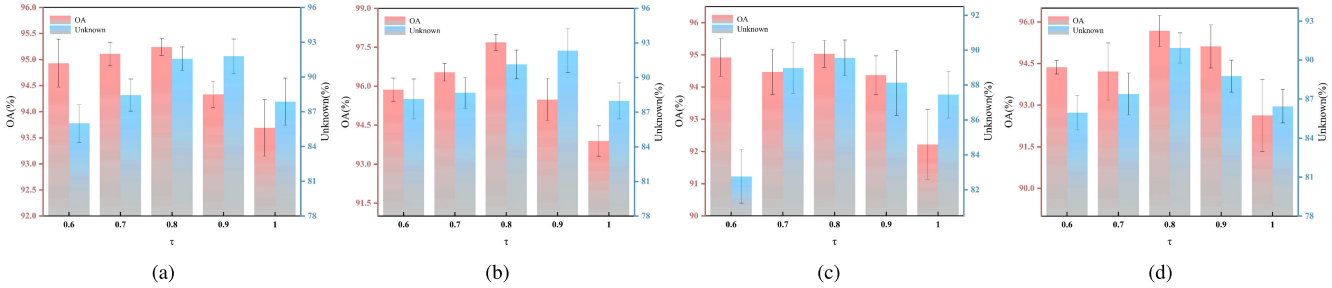


Fig. 16. Experimental results of different probability threshold τ on four datasets. (a) Salinas dataset. (b) DTL dataset. (c) LN02 dataset. (d) KSC dataset.

E. Parameter and Sensitivity Analysis

In this section, a further exploration and analysis are conducted on the parameters that potentially influence the model's performance, including the number of training samples N_{kn} for known categories, the order α of the WFrFT, the probability threshold τ and the number of unknown categories N_{un}. To evaluate the impact of these parameters, OA and classification accuracy of unknown categories are adopted as the evaluation metrics.

1) Number of Training Samples for Known Categories: To explore the impact of the number of training samples per known category N_{kn} on the classification performance of FrHSPL, experiments were conducted with N_{kn} set to 10, 20, 30, and 40, respectively. The experimental results, illustrated in Fig. 14, provide a comparison of FrHSPL against the closed-set method SSRN and three open-set methods across different numbers of training samples.

Observations reveal that as N_{kn} increases, the performance of various methods improves to varying degrees. Notably, FrHSPL consistently maintains superior classification capability, indicating that the HSPL strategy can proficiently leverage limited training data in modeling known category characteristics. In terms of unknown category recognition, certain open-set methods, such as SSLR and MDL4OW, exhibit noticeable instability when confronted with varying N_{kn}. In contrast, FrHSPL achieves a high level of accuracy in unknown category recognition, with relatively stable performance. Overall, FrHSPL effectively balances the classification performance of both known and unknown categories under varying numbers of training samples, showcasing its robustness in open-set scenarios.

2) Order α of WFrFT: The order of the WFrFT determines the degree of transformation of spatial–spectral features between the time and frequency domains. Typically, it adopts any value within the range of 0–1. In the proposed FrHSPL, the order α is designed as an adaptive parameter that can be learned and optimized concurrently with model training. The experimental results are presented in Fig. 15, where a_1–a_4 denote the optimal model performance achieved by order α after adaptive adjustment across four datasets.

In the Salians dataset, the OA gradually increases as α rises, peaking at 95.24% when the order adapts to a value of 0.77. Beyond this point, further increasing the order leads to a decline in performance. For the DTL dataset, as α increases, not only does the OA steadily rise but also the accuracy of the unknown category improves accordingly, both reaching their highest points at α of 0.98, with values of 97.69% and 91.13%, respectively. This indicates that the utilization of frequency-domain information is particularly crucial for enhancing model performance in this dataset. However, in the LN02 and KSC datasets, the optimal model performance is achieved at the order of 0.58 and 0.63, respectively, where a balance between frequency-domain information and spatial–spectral information may be achieved. In summary, the adaptive adjustment of the order α enables the model to exhibit optimal classification performance across different datasets. This also validates the rationality and effectiveness of dynamically adjusting the order based on the specific characteristics of the dataset and model parameters.

3) Probability Threshold τ: Considering the uncertainty associated with the logit distribution of unknown categories, the testing samples are grouped into a known-class set and an uncertain set. This is achieved by analyzing the difference between the maximum and second-maximum predicted probability within the closed set. OSR is subsequently conducted within the uncertain set.

Fig. 16 exhibits the specific impact of different probability threshold τ on model performance across four datasets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

TABLE X
EXPERIMENTAL RESULTS WITH DIFFERENT NUMBER OF UNKNOWN CATEGORIES ON SALINAS DATASETS

N_{un}		SSRN [17]	FrFSSPN [28]	MDL4OW [42]	SSLR [43]	**FrHSPL**
1	OA	84.68	94.95	88.08	90.80	95.24
	AA	90.92	96.56	93.03	92.31	96.88
	Kappa	83.31	94.76	89.52	90.20	94.89
	Unknown	80.69	87.63	80.86	55.58	91.56
2	OA	82.56	92.65	82.57	86.53	93.51
	AA	90.46	95.27	91.32	93.11	95.67
	Kappa	78.86	91.39	81.28	85.55	92.88
	Unknown	68.07	71.91	74.35	53.14	86.73
6	OA	74.09	81.61	72.93	73.55	82.38
	AA	88.24	90.45	90.38	91.05	91.22
	Kappa	70.56	78.74	69.39	71.22	81.23
	Unknown	53.37	56.40	58.25	36.69	65.98
9	OA	70.75	73.86	67.39	57.42	78.30
	AA	87.13	89.18	88.11	85.53	90.54
	Kappa	65.95	65.73	62.24	52.40	74.18
	Unknown	45.30	38.41	44.46	24.32	53.72
12	OA	53.55	63.92	38.49	36.47	72.73
	AA	86.49	87.25	83.71	81.49	90.92
	Kappa	43.62	48.60	32.98	32.36	59.34
	Unknown	32.26	40.19	38.48	22.06	56.46

TABLE XI
EXPERIMENTAL RESULTS WITH DIFFERENT NUMBER OF UNKNOWN CATEGORIES ON DTL DATASETS

N_{un}		SSRN [17]	FrFSSPN [28]	MDL4OW [42]	SSLR [43]	**FrHSPL**
1	OA	92.44	96.46	92.12	94.19	97.69
	AA	91.21	95.41	91.35	92.80	97.21
	Kappa	93.21	95.89	93.34	92.93	97.19
	Unknown	55.38	84.97	72.98	71.24	91.13
2	OA	88.41	86.59	87.29	88.39	95.62
	AA	89.57	86.21	87.91	89.21	95.07
	Kappa	85.54	80.86	83.94	85.24	95.04
	Unknown	52.64	75.84	67.97	55.46	88.04
3	OA	70.72	84.77	73.65	70.62	89.87
	AA	83.07	85.27	82.02	81.96	89.94
	Kappa	62.43	81.83	64.99	61.81	85.46
	Unknown	32.39	65.78	47.14	35.83	67.35
4	OA	49.95	58.85	75.58	77.68	75.55
	AA	72.84	72.27	71.24	86.87	74.71
	Kappa	34.71	44.15	73.91	75.94	63.49
	Unknown	18.41	26.88	78.80	51.29	48.86

When $\tau = 1$, it implies performing OSR directly on all test samples. However, the results indicate that neither the OA nor the recognition accuracy of unknown categories met the expected standards. This unsatisfactory performance is primarily attributed to a substantial number of known-class samples being erroneously identified as unknown. In addition, This also indicates that the high precision of unknown category recognition comes at the expense of sacrificing some accuracy in identifying known ones. With the incorporation of identify inference and the increase in threshold τ, the determination of the uncertain set becomes more stringent, leading to an improvement in the model performance. When τ reaches 0.8, the model exhibits relatively high performance. However, setting the threshold too high leads to more known categories being divided into the uncertain set, resulting in their incorrect classification as unknown. Overall, the impact of different probability thresholds τ on model performance is relatively minor. When τ is 0.8, the model achieves optimal performance in terms of OA and recognition precision for unknown categories.

4) Number of Unknown Categories: The discussion and analysis in the aforementioned experiments were conducted by designating a specific category within the dataset as an unknown class. To enhance the openness of the data and more comprehensively validate the conclusions, we augment the number of unknown categories while correspondingly decreasing the quantity of known categories. Based on the proportion of sample counts for a given land cover category, several categories with lower frequency are selected to serve as the unknown categories. The number of unknown categories is set

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
EXPERIMENTAL RESULTS WITH DIFFERENT NUMBER OF UNKNOWN CATEGORIES ON LN02 DATASETS

N_{un}		SSRN [17]	FrFSSPN [28]	MDL4OW [42]	SSLR [43]	**FrHSPL**
1	OA	93.80	93.89	93.83	94.56	95.03
	AA	87.83	92.87	87.75	87.23	94.43
	Kappa	93.22	93.07	92.32	93.90	94.75
	Unknown	57.11	86.68	76.11	52.51	89.56
2	OA	92.04	91.39	88.45	89.43	91.64
	AA	86.16	90.86	86.46	86.03	90.27
	Kappa	90.76	90.26	86.74	88.04	90.94
	Unknown	51.96	77.93	71.34	48.22	85.02
4	OA	87.49	88.48	72.85	79.49	90.11
	AA	82.26	86.76	80.97	82.52	87.29
	Kappa	84.46	87.35	69.66	76.73	89.08
	Unknown	30.86	64.27	59.44	45.47	72.39
5	OA	62.83	67.53	63.11	68.35	72.72
	AA	76.48	81.17	70.49	73.49	83.01
	Kappa	60.39	62.86	56.83	63.98	70.78
	Unknown	25.65	51.44	48.77	38.65	56.17
6	OA	47.44	64.54	48.74	61.20	64.98
	AA	73.17	80.77	67.63	70.73	75.45
	Kappa	41.18	56.15	38.48	56.07	58.74
	Unknown	17.71	40.84	42.31	33.50	44.57

TABLE XIII
EXPERIMENTAL RESULTS WITH DIFFERENT NUMBER OF UNKNOWN CATEGORIES ON KSC DATASETS

N_{un}		SSRN [17]	FrFSSPN [28]	MDL4OW [42]	SSLR [43]	**FrHSPL**
1	OA	88.46	93.15	92.84	91.34	95.68
	AA	89.02	94.64	90.12	86.01	95.44
	Kappa	87.27	92.33	92.40	90.38	95.63
	Unknown	62.69	44.69	23.90	15.64	90.93
2	OA	77.92	73.28	78.01	78.78	90.11
	AA	84.82	90.66	85.51	90.49	83.48
	Kappa	74.92	70.68	74.94	78.13	88.37
	Unknown	31.88	20.65	11.69	30.82	89.73
5	OA	50.71	46.53	45.29	70.31	70.77
	AA	81.75	85.82	82.37	89.05	57.53
	Kappa	44.42	42.72	61.70	63.25	58.82
	Unknown	27.01	7.62	2.37	41.23	80.63
7	OA	47.39	36.49	43.11	62.75	81.69
	AA	75.71	81.57	79.14	86.34	79.25
	Kappa	38.98	31.97	54.29	51.52	75.59
	Unknown	16.68	0.00	14.72	42.85	82.05
9	OA	35.98	29.05	54.86	51.66	66.91
	AA	71.48	77.00	82.68	82.59	81.66
	Kappa	26.43	24.11	55.22	37.78	54.49
	Unknown	11.67	0.00	28.09	32.44	56.90

to {1, 2, 6, 9, 12}, {1, 2, 3, 4}, {1, 2, 4, 5, 6}, and {1, 2, 5, 7, 9} for Salinas, DTL, LN02, and KSC, respectively.

As observed from Tables X to XIII, with an increasing number of unknown categories, the performance of all methods, in terms of both OA and the accuracy of unknown categories, inevitably deteriorates. This indicates that the greater the interference from unknown categories, the more challenging the classification task becomes. Notably, FrHSPL maintains a high-classification performance, with a relatively small decline in accuracy, suggesting its effectiveness in balancing the classification of known and unknown categories. Especially, when the number of unknown categories approaches or even exceeds that of known categories, the performance of other methods, such as SSRN and MDL4OW, degrades significantly. In contrast, FrHSPL retains a strong discriminative capability, even in scenarios with a large number of unknown categories. Conclusively, FrHSPL exhibits exceptional adaptability and maintains superior classification performance across varying numbers of unknown categories.

F. Analysis of Computational Complexity

To evaluate the computational complexity of FrHSPL, Table XIV provides the total number of trainable parameters

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

TABLE XIV
PARAMETERS AND COMPUTATION TIME (SECONDS) ACROSS DIFFERENT METHODS

		SSRN [17]	SSFTT [21]	DBDA [24]	FrFSSPN [28]	MDL4OW [42]	SSLR [43]	SSMTL [44]	**FrHSPL**
Salinas	Training time(s)	1101.65	16.14	416.15	43.62	251.45	708.46	2110.68	46.41
	Testing time(s)	20.63	3.22	66.09	21.22	81.97	101.07	213.78	24.83
	Params(M)	1.41	0.57	1.49	4.47	0.43	0.73	1.77	3.46
DTL	Training time(s)	361.09	4.67	124.89	32.54	78.71	245.19	1236.03	39.36
	Testing time(s)	42.43	3.88	77.95	43.78	123.66	141.26	317.64	34.31
	Params(M)	0.53	0.17	0.58	8.55	0.55	0.46	1.72	6.87
LN02	Training time(s)	349.92	7.35	117.35	31.31	119.62	305.55	1237.18	32.45
	Testing time(s)	33.39	6.71	105.78	42.29	87.55	140.77	712.83	35.62
	Params(M)	1.06	0.56	1.07	2.72	0.36	0.31	1.72	1.63
KSC	Training time(s)	581.44	7.69	118.66	35.36	191.24	472.26	1969.27	36.37
	Testing time(s)	43.19	7.48	81.19	52.65	168.63	200.62	768.96	39.62
	Params(M)	0.33	0.15	0.38	3.58	0.39	0.34	1.72	2.49

and the runtime of FrHSPL alongside other comparison methods.

As evident from the data presented in the table, SSFTT within the closed-set methods exhibits the lowest computation time consumption. In contrast, SSMTL incurs substantially higher training and testing times due to the complexity of its pretraining procedures. In addition, MDL4OW and SSLR lead to additional computational overhead, as the identification of unknown categories depends on the reconstruction error of each pixel. Although FrHSPL introduces an increase in parameters to a certain extent by incorporating the WFrFT, it does not result in an excessive growth in computation time. Overall, FrHSPL achieves optimal performance within a reasonable runtime.

IV. CONCLUSION

In this article, a novel FrHSPL method is proposed for open-set HSIC. To optimize the exploitation of metric space, the HSPL strategy is designed to constrain the features of known categories so that they are uniformly distributed on a hypersphere. Therefore, HSPL can effectively enhance interclass separability while maintaining intraclass compactness. Meanwhile, to alleviate the challenge of indistinguishable spectral differences, the FSSIA module is devised to elegantly aggregate fractional domain information with spatial–spectral information. A dual-branch feature extraction network is also designed to ensure the efficient utilization of frequency-spatial–spectral information. Furthermore, the OSR module is designed grouped on the prototypes of each known category and their respective prototype radii. Fully exploring and leveraging the distribution characteristics of known categories can enhance the model's capability to identify unknown categories. Compared with the current three closed-set and four open-set HSIC methods, FrHSPL exhibits superior classification performance across four benchmark datasets.

REFERENCES

[1] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019.

[2] C. Zhao et al., "Hyperspectral image classification with multi-attention transformer and adaptive superpixel segmentation-based active learning," *IEEE Trans. Image Process.*, vol. 32, pp. 3606–3621, 2023.

[3] Y. Du, X. Li, L. Shi, F. Li, and T. Xu, "A prototype network for hyperspectral image open-set classification based on feature invariance and weighted Pearson distance measurement," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5506917.

[4] Z. Xie, J. Hu, X. Kang, P. Duan, and S. Li, "Multilayer global spectral–spatial attention network for wetland hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518913.

[5] F. Guo et al., "Multisource feature embedding and interaction fusion network for coastal wetland classification with hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5509516.

[6] C. Zhao, C. Li, S. Feng, and W. Li, "Spectral–spatial anomaly detection via collaborative representation constraint stacked autoencoders for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[7] X. Zhao, R. Tao, W. Li, W. Philips, and W. Liao, "Fractional Gabor convolutional network for multisource remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503818.

[8] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[9] N. Li, S. Jiang, J. Xue, S. Ye, and S. Jia, "Texture-aware self-attention model for hyperspectral tree species classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502215.

[10] B. Rasti et al., "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[11] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[12] J. Xia, J. Chanussot, P. Du, and X. He, "Rotation-based support vector machine ensemble in classification of hyperspectral data with limited training samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1519–1531, Mar. 2016.

[13] L. Samaniego, A. Bardossy, and K. Schulz, "Supervised classification of remotely sensed imagery using a modified k-NN technique," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2112–2125, Jul. 2008.

[14] Y. E. SahIn, S. Arisoy, and K. Kayabol, "Anomaly detection with Bayesian Gauss background model in hyperspectral images," in *Proc. 26th Signal Process. Commun. Appl. Conf. (SIU)*, May 2018, pp. 1–4.

[15] C. Yu et al., "Distillation-constrained prototype representation network for hyperspectral image incremental classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507414.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.

[16] K. Chhapariya, A. Benoit, K. M. Buddhiraju, and A. Kumar, "A multitask deep learning model for classification and regression of hyperspectral images: Application to a large-scale dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5510220.

[17] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[18] H. Yu, Z. Xu, K. Zheng, D. Hong, H. Yang, and M. Song, "MSTNet: A multilevel spectral–spatial transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532513.

[19] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[20] H. Feng, Y. Wang, C. Chen, D. Xu, Z. Zhao, and T. Zhao, "Hyperspectral image classification framework based on multichannel graph convolutional networks and class-guided attention mechanism," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5515115.

[21] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[22] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[23] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020. [Online]. Available: https://www.mdpi.com/2072-4292/12/3/582

[24] C. Fu, B. Du, and L. Zhang, "ReSC-net: Hyperspectral image classification based on attention-enhanced residual module and spatial-channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518615.

[25] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[26] Y. Hu, J. Zhang, Y. Ma, J. An, G. Ren, and X. Li, "Hyperspectral coastal wetland classification based on a multiobject convolutional neural network model and decision fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1110–1114, Jul. 2019.

[27] J. Luo, Z. He, H. Lin, and H. Wu, "Biscale convolutional self-attention network for hyperspectral coastal wetlands classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[28] M. Chen et al., "Fractional Fourier-based frequency-spatial–spectral prototype network for agricultural hyperspectral image open-set classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514014.

[29] R. Tao, X. Zhao, W. Li, H.-C. Li, and Q. Du, "Hyperspectral anomaly detection by fractional Fourier entropy," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 4920–4929, Dec. 2019.

[30] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 2314–2326, Feb. 2024.

[31] L. Zhang, B. Cheng, S. Tan, and Y. Wang, "Fractional Fourier transform based joint adaptive subspace detection for hyperspectral anomaly detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[32] Z.-G. Liu, Y.-M. Fu, Q. Pan, and Z.-W. Zhang, "Orientational distribution learning with hierarchical spatial attention for open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8757–8772, Jul. 2023.

[33] H. Sun et al., "Deep feature reconstruction learning for open-set classification of remote-sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[34] J. Peng, W. Sun, and Q. Du, "Self-paced joint sparse representation for the classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1183–1194, Feb. 2019.

[35] Z. Liu, G. Chen, Z. Li, Y. Kang, S. Qu, and C. Jiang, "PSDC: A prototype-based shared-dummy classifier model for open-set domain adaptation," *IEEE Trans. Cybern.*, vol. 53, no. 11, pp. 7353–7366, Nov. 2023.

[36] Z. Xie, P. Duan, W. Liu, X. Kang, X. Wei, and S. Li, "Feature consistency-based prototype network for open-set hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9286–9296, Jul. 2024.

[37] Z. Li, K. Bi, Y. Wang, Z. Fang, and J. Zhang, "Supervised contrastive learning for open-set hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[38] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, and T. E. Boult, "Toward open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 7, pp. 1757–1772, Jul. 2013.

[39] Z. Qin, L. Yang, F. Gao, Q. Hu, and C. Shen, "Uncertainty-aware aggregation for federated open set domain adaptation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 7548–7562, Jun. 2024.

[40] X. Yin, B. Cao, Q. Hu, and Q. Wang, "RD-OpenMax: Rethinking OpenMax for robust realistic open-set recognition," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 7565–7579, Apr. 2025.

[41] H. Huang, Y. Wang, Q. Hu, and M.-M. Cheng, "Class-specific semantic reconstruction for open set recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4214–4228, Apr. 2023.

[42] S. Liu, Q. Shi, and L. Zhang, "Few-shot hyperspectral image classification with unknown classes using multitask deep learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5085–5102, Jun. 2021.

[43] J. Yue, L. Fang, and M. He, "Spectral–spatial latent reconstruction for open-set hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5227–5241, 2022.

[44] C. Mu, Y. Liu, X. Yan, A. Ali, and Y. Liu, "Few-shot open-set hyperspectral image classification with adaptive threshold using self-supervised multitask learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5526618.

[45] F. Ji et al., "Spectral–spatial evidential learning network for open-set hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5503617.

[46] Y. Sun, B. Liu, R. Wang, P. Zhang, and M. Dai, "Spectral–spatial MLP-like network with reciprocal points learning for open-set hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513218.

[47] Z. Wang, L. Mei, X. Sha, and N. Zhang, "BER analysis of STBC hybrid carrier system based on WFRFT with frequency domain equalization," *Sci. China Inf. Sci.*, vol. 61, no. 8, pp. 1–13, Aug. 2018.

[48] M. S. E. Saadabadi, A. Dabouei, S. R. Malakshan, and N. M. Nasrabadi, "Hyperspherical classification with dynamic label-to-prototype assignment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 17333–17342.

[49] B. Qin et al., "Hyperspherical structural-aware distillation enhanced spatial–spectral bidirectional interaction network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524714.

[50] H. Yu et al., "Global spatial and local spectral similarity-based manifold learning group sparse representation for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3043–3056, May 2020.

[51] B. Tu, W. He, Q. Li, Y. Peng, and A. Plaza, "A new context-aware framework for defending against adversarial attacks in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505114.

[52] H. Yu, Z. Ling, K. Zheng, L. Gao, J. Li, and J. Chanussot, "Unsupervised hyperspectral and multispectral image fusion with deep spectral–spatial collaborative constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534114.

[53] Z. Gong, P. Zhong, Y. Yu, W. Hu, and S. Li, "A CNN with multiscale convolution and diversified metric for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3599–3618, Jun. 2019.

[54] A. Bendale and T. E. Boult, "Towards open set deep networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1563–1572.

[55] C. Geng, S.-J. Huang, and S. Chen, "Recent advances in open set recognition: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3614–3631, Oct. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 23:19:08 UTC from IEEE Xplore. Restrictions apply.